CHARUSAT University

Subject: Computer networks Total Marks: 100

Section - I: Multiple Choice Questions

1. In a network, a client requests a large file from a server using TCP. The server breaks the file into smaller segments and sends them to the client. However, during transmission, packet 3 is lost in the network. Which of the following statements is true?

A) The client will receive packets 1, 2, and 4, but not packet 3.

B) The client will receive packets 1 and 2, and then wait for packet 3 to be retransmitted.

C) The client will receive packets 1, 2, and 4, and then request retransmission of packet 3.

D) The client will not receive any packets until packet 3 is retransmitted.

2. What is the primary reason why a web browser uses TCP for requesting a web page from a server, but UDP for DNS lookups?

A) TCP is more reliable for large data transfers, while UDP is better suited for small packets.

B) TCP is required for HTTP, while UDP is required for DNS.

C) TCP is used for connection-oriented communication, whereas UDP is used for connectionless communication, and DNS lookups require a faster response time.

D) TCP is used for client-server communication, whereas UDP is used for peer-to-peer communication.

3. In a network, a server is configured to simultaneously support both TCP and UDP for a specific application. If the application requires guaranteed delivery of packets in a specific order, but also needs to minimize latency and jitter, which of the following configurations is most likely to be used?

A) TCP for signaling and UDP for data transfer

B) UDP for signaling and TCP for data transfer

C) TCP for both signaling and data transfer

D) UDP for both signaling and data transfer

4. In a network, a client requests a large file from a server using TCP. During the transmission, a packet is lost due to network congestion. Which of the following statements is true about the retransmission of the lost packet?

A) The client will retransmit the entire file from the beginning.

B) The server will retransmit the entire file from the beginning.

C) The server will retransmit only the lost packet after receiving a request from the client.

D) The server will retransmit all packets sent after the lost packet.

5. In a network, a client and a server are communicating using a connection-oriented protocol. The client sends a large file to the server, but due to network congestion, one of the packets is lost during transmission. How will the client and server react to this situation?

A) The client will resend the lost packet, and the server will reassemble the file in the correct order.

B) The client will resend the entire file, and the server will discard the previously received packets.

C) The server will request the retransmission of the lost packet, and the client will acknowledge the request.

D) The client and server will terminate the connection, and the file transfer will be aborted.

6. What is the primary reason why UDP is preferred over TCP for real-time applications, despite UDP's lack of guaranteed delivery and packet ordering?

A) UDP has lower overhead due to the absence of connection establishment and teardown.

B) TCP's congestion avoidance algorithm introduces variable delays, which are unacceptable in real-time applications.

C) UDP's connectionless nature allows it to handle packet loss more efficiently than TCP.

D) UDP's error-checking mechanism is more robust than TCP's, ensuring accurate data transmission.

7. In a network, a server is designed to handle a large volume of concurrent requests from clients. To ensure reliable communication, the server uses a connection-oriented protocol. However, the clients are experiencing high latency due to the overhead of establishing and terminating connections. Which of the following modifications to the server's communication protocol would likely minimize latency while maintaining reliability?

A) Implementing UDP with additional error-checking mechanisms

B) Using TCP with a smaller window size to reduce congestion

C) Employing a connectionless protocol with built-in reliability features

D) Implementing a protocol that combines the connection establishment of TCP with the connectionless nature of UDP

8. Which of the following scenarios would most likely benefit from using UDP over TCP?

A) File transfer between two systems where guaranteed delivery is crucial

B) Real-time video streaming where occasional packet loss is acceptable

C) Remote access to a database where every query and response must be received in the correct order

D) Online banking transaction where data integrity and guaranteed delivery are essential

9. What is the primary reason why a web browser using TCP may experience slower performance when retrieving multiple small files from a web server, compared to using UDP?

A) TCP's connection establishment overhead dominates the transfer time for small files.

B) TCP's error-correction mechanism introduces additional latency.

C) The browser's DNS resolution takes longer for TCP than for UDP.

D) UDP's packet sequencing is more efficient for small files.

10. In a network where real-time video streaming is taking place, the application layer is configured to use UDP as the transport layer protocol. Which of the following is a consequence of this design decision?

A) The video streaming application will experience increased latency due to UDP's lack of error correction mechanisms.

B) The network will be more prone to congestion collapse due to UDP's inability to throttle back transmission rates.

C) The video streaming application will be more resilient to packet losses, as UDP will retransmit lost packets.

D) The video streaming application will be able to guarantee a specific Quality of Service (QoS) for its traffic.

11. What is the primary reason why TCP is preferred over UDP for file transfers in a network, despite UDP's lower overhead and faster transmission rate?

A) TCP provides error detection and correction, while UDP does not

B) TCP is better suited for real-time applications, such as video streaming

C) TCP is more resistant to network congestion, whereas UDP is not

D) TCP is more widely supported by network devices and operating systems

12. Which of the following scenarios would most likely benefit from using UDP as the transport layer protocol over TCP?

A) A real-time video conferencing application that can tolerate some packet loss but requires low latency.

B) A file transfer protocol that requires guaranteed delivery of files in the correct order.

C) A web server that needs to establish multiple concurrent connections with clients.

D) A networked multiplayer game that requires reliable, in-order delivery of game state updates.

13. Which of the following scenarios would most likely benefit from using UDP as the transport layer protocol instead of TCP?

A) A file transfer protocol that requires guaranteed delivery of files

B) A real-time video streaming application that can tolerate some packet loss

C) A web browsing application that requires sequential delivery of HTML pages

D) A remote access VPN that requires guaranteed encryption and authentication

14. What is the primary reason why UDP is preferred over TCP for live video streaming applications, despite the reliability issues associated with UDP?

A) TCP's connection establishment and teardown processes introduce significant latency.

B) UDP's connectionless nature allows for better handling of packet loss and reordering.

C) TCP's congestion control mechanism can cause significant throughput fluctuations.

D) UDP's error-checking mechanism is more efficient than TCP's error-correction mechanism.

15. What is the primary reason why TCP is often preferred over UDP for file transfer applications, despite being slower?

A) TCP provides better error detection and correction mechanisms

B) TCP allows for faster packet transmission due to its connectionless nature

C) UDP is more vulnerable to packet sniffing and interception

D) TCP enables packet multiplexing, allowing multiple files to be transferred simultaneously

16. Which of the following statements is true about the relationship between TCP and UDP in a network where real-time data transmission is critical?

A) Since UDP is connectionless, it is more reliable than TCP for real-time applications.

B) TCP's error-checking mechanism makes it more suitable for real-time data transmission than UDP.

C) UDP's lack of error-checking mechanism makes it more suitable for real-time data transmission than TCP.

D) Both TCP and UDP are equally suitable for real-time data transmission, and the choice between them is arbitrary.

17. In a network, a client sends a large file to a server using a connection-oriented protocol. During the transmission, a packet is lost due to network congestion. Which of the following statements is true about the retransmission of the lost packet?

A) The client will retransmit the entire file from the beginning.

B) The server will request the retransmission of the lost packet using an acknowledgement packet.

C) The client will retransmit the lost packet only after receiving a timeout signal.

D) The retransmission of the lost packet is not possible in a connection-oriented protocol.

18. What is the primary reason why a web browser uses TCP for establishing a connection to a web server, but online gaming applications often use UDP?

A) TCP is more secure than UDP

B) TCP provides guaranteed delivery of packets, whereas UDP does not

C) UDP has lower latency than TCP due to its connectionless nature

D) TCP is better suited for bulk data transfer, whereas UDP is better for real-time data transfer

19. What is the primary reason why UDP is preferred over TCP for real-time applications such as video conferencing, despite UDP's lack of guarantee for packet delivery?

A) UDP has lower overhead in terms of header size

B) TCP's congestion control mechanism can lead to variable latency

C) UDP is better suited for packet-based transmissions

D) TCP's connection establishment process is too slow for real-time applications

20. In a network where real-time video conferencing is taking place, the application layer is using UDP as the transport layer protocol. If the packets arrive out of order, which of the following is NOT a reason why the receiver's application would not be severely impacted?

A) UDP does not provide sequencing information.

B) The receiver's application can request retransmission of the packets.

C) The video conferencing application can tolerate some packet loss.

D) The receiver's application can rearrange the packets based on timestamp information.

Section - II: Descriptive / Case Study-Based Questions

1. A large-scale e-commerce company, "ShopEazy," has recently expanded its operations to include live video streaming of product demonstrations and customer support sessions. The company's IT department has noticed a significant increase in network congestion, resulting in frequent packet losses and delayed video streams. The video streaming application uses UDP as the transport-layer protocol, while the customer support chat system relies on TCP. Analyze the possible causes of network congestion and propose a solution to optimize the video streaming and customer support services. Consider the characteristics of TCP and UDP, their suitability for different applications, and the potential impact of packet loss and delay on user experience.

2. A popular online gaming company, "GameFever", is experiencing issues with its real-time multiplayer game, "Epic Quest". The game utilizes TCP for communication between the client and server, but players are complaining about lag and disconnections during intense battle scenes. The game developers have identified that the problem arises when multiple players cast spells simultaneously, causing a sudden surge in network traffic. Given that the game requires low latency and high throughput, but also guarantees data integrity, propose a solution to mitigate the issue. Consider the trade-offs between using TCP and UDP, and discuss the potential implications of your proposed solution on the game's performance and player experience.

3. A popular online gaming company, "GameOn", has developed a new multiplayer game that allows players to engage in real-time battles across the globe. The game uses a combination of TCP and UDP protocols to ensure seamless communication between players. However, the company's network engineers have noticed that during peak hours, a significant number of players are experiencing lag and disconnections, resulting in a poor gaming experience. Upon further investigation, the engineers have discovered that the game's server is sending a large amount of data using TCP, including game state updates, player positions, and chat messages. Meanwhile, the client-side is using UDP to transmit player inputs, such as movement and attack commands. Analyzing the network traffic, the engineers have observed that during peak hours, the TCP connection is experiencing high latency and packet loss, causing the game state to become desynchronized among players. However, the UDP traffic appears to be unaffected, with packets being delivered quickly and efficiently. What adjustments would you recommend to the network engineers to optimize the communication protocols and improve the overall gaming experience, taking into account the trade-offs between reliability, latency, and throughput in TCP and UDP? Be sure to justify your suggestions with clear explanations of how they address the specific issues observed in the network traffic.

4. A leading e-commerce company is experiencing issues with its online payment processing system. The system uses a combination of TCP and UDP protocols to handle transactions. However, the company's IT department has observed that a significant number of transactions are being lost during peak hours, resulting in revenue loss and customer dissatisfaction. Upon further investigation, the IT department discovers that the system is configured to use TCP for payment processing and UDP for sending order confirmations to customers. The network infrastructure is robust, with high-speed switches and routers, and

the servers are capable of handling high volumes of traffic. Analyzing the network traffic, the IT department finds that the TCP connections are experiencing high latency and packet loss during peak hours, while the UDP packets are being delivered quickly and efficiently. However, the UDP packets are not guaranteed to reach the customers, and the company's customer service team is receiving complaints about missing order confirmations. What changes would you recommend to the company's IT department to improve the reliability and efficiency of the online payment processing system, considering the trade-offs between TCP and UDP protocols? Be sure to justify your recommendations with technical explanations and potential implications for the system's performance.

5. A popular online gaming company is launching a new multiplayer game that requires real-time communication between players. The game involves fast-paced action and requires rapid updates of player positions, scores, and game state. The company's network engineers are debating whether to use TCP or UDP as the transport-layer protocol for the game's communication. The game's requirements are as follows: - The game client and server must exchange packets at a rate of at least 10 packets per second to ensure a seamless gaming experience. - The average packet size is 128 bytes. - The network infrastructure is a high-speed, low-latency LAN with an average round-trip time of 20 milliseconds. - The game can tolerate some packet loss, but retransmission of lost packets may introduce unacceptable latency. - The game's server can handle up to 1000 concurrent connections. Considering these requirements, explain the advantages and disadvantages of using TCP versus UDP as the transport-layer protocol for the game's communication. Which protocol would you recommend and why? Be sure to justify your answer with specific examples and technical details.

6. Error: 429 Client Error: Too Many Requests for url: https://api.groq.com/openai/v1/chat/completions

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